

# Segmentation of cars with smoke

Dageng Ding

*School of Electrical Engineering and Computer Science, University of Ottawa*  
800 King Edward Avenue, Ottawa, Canada, K1N 6N5  
dding042@uottawa.ca

**Abstract**—An issue is found that the model is unable to segment the cars covered by smoke properly. Several experiments are made to solve this issue, which includes model evaluating and training dataset augmentation. To be more specific, we tested how different combination of models (architecture + backbone) performs under the same situation, trying to find a model combination that could best avoid the improper segment issue. We then rebuild the original training dataset, with a more targeted dataset that aims to provide smoke covered car images. We tried different versions of rebuilt training datasets for comparison. Eventually, we have chosen the segmentation model and training dataset to solve the issue.

**Index Terms**—object segmentation, mobileone, dataset modification, depth-wise separable convolution

## I. INTRODUCTION

Object segmentation is a technology of splitting different objects into segments by their classes for images. In our application of object segmentation, we are attempting to segment car objects from a video (which can be seen as a series of images for object segmentation). However, it is observed that whenever the car is covered by smoke, the current model (for object segmentation) is unable to identify or segment the car properly. To be more specific, as shown in Fig. 1, the model performs well at first while the car is not covered by smoke. But later in the video, as shown in Fig. 2, as the smoke starts to cover partially of the car, the model fails to segment or even identify it as a car object. Therefore, the current model must be changed to solve this issue (“Car vanishing issue” in this report ,referring to the issue that a model is unable to segment or identify smoke-covered car).

In order to solve the issue, we start by finding a new model (including architecture, backbone and encoder) that would perform better against smoke under the same dataset configuration. And then, retrain the model with a modified dataset that is targeted to segment smoke-covered car objects.

Execution time is also one important requirement for us since our final goal is to provide object segmentation through live video processing. In this case, the new model should be small and fast instead of complex and slow.

## II. TESTING WITH NEW MODELS

Our current original (which is initially used and found out the car vanishing issue) is as shown in following: “mobilenet\_v3\_large” [1] as the pretrained backbone and

“LRASPP\_MobileNet\_V3\_Large\_Weights.COCO\_WITH\_VOC\_LABELS\_V1” as the pretrained weights. The



Fig. 1: Original frame before segmentation where there is no smoke covering the racing car (left) and the frame after segmentation (right).



Fig. 2: Original frame before segmentation where there is smoke covering the racing car (left) and the frame after segmentation (right).

mobilenet is a light-weight model aiming to trade off accuracy for lower latency [1]. We decide to use a bigger model deeplabv3 [2] for experiments, to test if using a more complex model could solve the vanishing car issue.

We experiment with some combinations as following:

| deeplab_version | backbone     | pretrained_Dataset | Car vanishing issue |
|-----------------|--------------|--------------------|---------------------|
| deeplabv3       | mobilenet    | voc [3]            | Exist               |
| deeplabv3_plus  | mobilenet    | cityscapes [4]     | Exist but less      |
| deeplabv3_plus  | mobilenet    | voc                | Exist               |
| deeplabv3       | ResNet50 [5] | voc                | Exist and worse     |
| deeplabv3_plus  | ResNet50     | voc                | Exist               |
| deeplabv3       | ResNet101    | voc                | Exist and worse     |
| deeplabv3_plus  | ResNet101    | voc                | Exist               |

TABLE I: Deeplabv3 with various backbone

After testing some combinations of deeplabv3 and various backbone, it is seen that the vanishing car problem still exists and sometimes gets even worse. Since using a more complex

model is not helping (and also increase time consumption), we then decide to try out several combinations of architecture and backbone (encoder + decoder) as follows:

- Unet [6] + ResNet34
- Unet + timm-regnetx\_040 [7]
- Unet + efficientnet-b4 [8]
- Unet + timm-efficientnet-b4
- Unet + mobileone\_s4 [9]
- Linknet [10] + ResNet34
- Linknet + timm-regnetx\_040
- Linknet + efficientnet-b4
- Linknet + timm-efficientnet-b4
- Linknet + timm-mobilenetv3\_large\_100

Besides all the combinations that are mentioned above, we also tested other architecture like FPN [11] and MAnet [12]. But we did not do any further test due to their poor performance. From the above combination, the “Unet + timm-regnetx\_040”, “Unet + efficientnet-b4”, “Unet + mobileone\_s4” and “Linknet+efficientnet-b4” (arranged from best to worst performance) seems to be performing well, decreasing the severity of car vanishing issue. Later in the experiment, the above combinations are used to test performance after rebuilding the new targeted dataset.

By now, we have several models or architecture combinations to be experimented on, it is time to rebuild a training dataset to retrain the model.

### III. DATASET AUGMENTATION

#### A. Addition of smoke

It is known that the performance of a model is determined by the training dataset provided to the model. And if the pretrained dataset does not contain any cars covered by the smoke or other smoke-like noise, it is likely that the model will fail to identify cars covered by smoke as it lacks of the corresponding knowledge (that is, training images).

After looking through the original training dataset and finding that there are no images of cars covered by smoke. In order to provide a more targeted dataset for car with smoke, we decide to add some smoke to the original training dataset. To be more specific, we are looking for the center of cars in the training dataset by looking at its labeled ground truth (i.e: finding its center of the label) and then overlapping with a smoke picture (with adjusted transparency to make both the smoke and car visible) into the center of car with a smoke ratio. The smoke ratio is defined as the ratio between the size of smoke that will be covering the car and the size of car. For example, a ratio of 1.0 means the smoke shall roughly cover the exact size of the car. Note that if there are multiple cars in the same image, the size of smoke would vary for each car depending on the size of the car and smoke ratio.

If the smoke-covered training images could cause the car vanishing problem (when use these images for object segmentation testing), then use them for training would help the model learn what a smoke-covered car looks like and possibly preventing the car vanishing issues. Based on this idea, a small

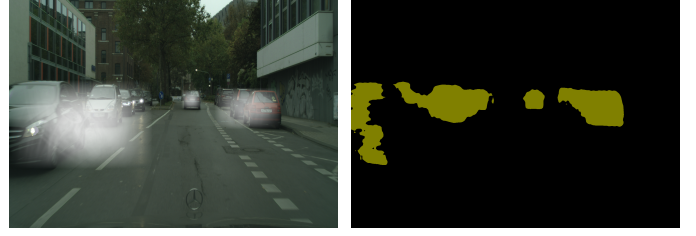


Fig. 3: Original frame with smoke added (left) and the frame after segmentation (right). The car to the bottom left is starting to vanish partially.



Fig. 4: Original frame with smoke covered the car (left) and the frame after segmentation (right). The smoke on the left of the car is segmented as a car object.

test is made to see if the pictures with smoke are enough to disrupt the model (using the original model “mobilenet\_v3” for testing). As shown in Fig. 3, it is seen that the smoke successfully disrupted the model from segmenting the car object properly, stating that pictures with smoke can cause the vanishing car issue and hereby could be used for the training dataset.

To summarize, addition of smoke to the dataset is done by overlapping an image of smoke onto the original dataset’s training images, to rebuild a dataset that is focused on car objects covered by smoke or smoke-liked noise.

#### B. Adjust the training dataset

Initially, the smoke ratio of 1.2 is used to modify the training dataset (by replacing the original dataset with new dataset), adding transparent smoke which is roughly 1.2 times the size of car to the center of cars. After experimenting with this new dataset (using model Unet + mobileone\_s4), as we can see in Fig. 4. The car with smoke is recognized by the model. But in the mean time, it is observed that the surrounding smoke is recognized as a part of the car.

The result shows that the addition of smoke is useful for solving the vanishing car issue, but the choice of model and smoke ratio needs to be adjusted to better off the performance as well as avoid the model from segmenting the surrounding smoke as a car object.

Before exploring more about the choice of smoke ratio and model, there is one more problem needs to be solved: How to replace the original training dataset with our new training dataset? As what we did in section III, we generate a new training dataset with pictures that are covered by smoke. We have 3 options to decide:

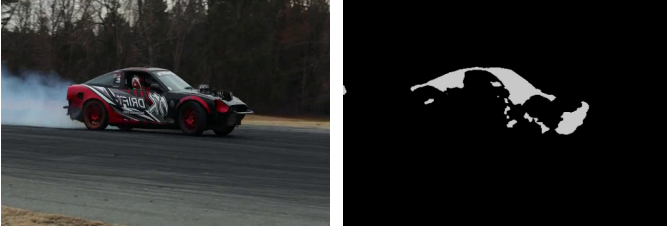


Fig. 5: Original frame with car not covered by smoke (left) and the frame after segmentation (right). Note a very major loss in the middle.

1. replace all of the original training images with smoke covered images
2. replace only a portion of the original training images with smoke covered images
3. expand the original training images with smoke covered images (that is, a training dataset containing both the smoke covered and original version of images)

The above 3 options are then tested using the same model (Unet + mobileone\_s4) within same seed, as described below with the specific actions:

- Fully smoke: replacing the original training images with smoke covered training images (that is, delete all the original images and replace it with smoke covered images)
- Mixed smoke: replacing only  $\sim 75\%$  (of total number of) original training images with smoke covered training images (the replacement is chosen randomly)
- Double smoke: expanding the original training images with our new training images (that is, keep the original images and paste the smoke covered images into the dataset)

We obtained the following results:

- Fully smoke: A new issue has raised as shown in Fig. 5. The model is unable to properly recognize and segment the car in normal scenario when the car is not yet covered by the smoke. However, the model is capable of recognizing the car while it is covered by the smoke. It is still unable to segment it properly (that is, sometimes, it segments the surrounding smoke as a part of the car)
- Mixed smoke: Similar issue of improper segmentation as shown in Fig. 5 happens to the model as well, while the overall performance is slightly better than the previous option.
- Double smoke: Unfortunately, the issue in Fig. 5 also happens to this option, but the affected number of frames and affected area of improper segmentation is slightly lower than the previous 2 options. Since its overall performance for segmenting cars (within/outside smoke) is the best among 3 options, it is then used as the default training dataset for the rest of the experiments.

However, even the double smoke is the best option among them all, there are still some problems to be solved later in the experiments (after adapting the double smoke method):

1. The improper segmentation while the car is not covered by the smoke, as seen in Fig. 5 (As later called as “non-smoke car vanishing issue”).
2. The car vanishing issue.

#### IV. FURTHER TESTING ON MODEL WITH SMALLER DATASET

It is ideal to use big dataset like cityscape to retrain the model. Considering about the time needed to retrain the model and the number of possible model combinations to be tested, it is best to find a smaller dataset to be tested on. Under this circumstances, a dataset named CamVid is chosen, as it only has  $\sim 370$  images containing car objects, which is perfect for making quick experiments on model performance.

After applying double smoke modification method to the CamVid dataset, we then choose the candidate models (from Section II) as well as several other potential models to be experimented as shown in the following:

- Linknet + efficientnet-b4
- Linknet + timm-regnetx\_040
- Unet + mobileone\_s4
- Unet + resnet34
- Unet + timm-regnety\_040
- Unet + timm-regnetx\_040
- Unet + efficientnet-b4

It is found that the “Unet + mobileone\_s4” has the best performance overall, with the slightest car vanishing issue. The “Unet + timm-regnety\_040” also has a good performance but performs worse than the “Unet + mobileone\_s4” in the non-smoke scenario.

Therefore, we are choosing the “Unet + mobileone\_s4” for the later experiments.

Instead of using dice loss, we are using a new metric called “IOU” to assess the model’s accuracy. It is defined by the following equation:

$$IOU = \frac{|GT \cap PD|}{|GT| \cup |PD|} \quad (1)$$

where GT is the area of groundtruth, PD is the area of prediction. It can help the model to solve the issue that counts car’s surrounding smoke as car object. Since if the issue occurs, the IOU score would be high, telling the model it is not optimal and needs to be adjusted in next iteration.

Note that IOU score is mainly used to reduce the issue, rather than a metric of measuring performance. For example, the final score of “Unet + timm-regnety\_040” could be 0.83 but its real performance is still worse off than the “Unet + mobileone\_s4” which has an IOU of 0.80.

#### V. MOVING TO BIG DATASET AND FINAL ADJUSTMENT

After locking on the Unet + mobileone model, the training dataset is then switched into a bigger dataset called cityscape (which contains  $\sim 3000$  images). As we previously mentioned, the smoke ratio is an important role on influencing the performance and should be put into use. Due to the time consuming



Fig. 6: Original frame with car fully covered by smoke (left) and the frame after segmentation (right). The performance is not very satisfactory.



Fig. 7: Original frame with car fully covered by smoke (left) and the frame after segmentation (right). After some image processing, the noise of improper labeling can be dealt with.

issue of huge dataset, we only chose 2 ratios (0.75 and 1.4) for the final adjustment.

The advantage and disadvantage of smoke ratio of 0.75 are pretty straight forward. As shown in Fig. 6, we can see the vanishing car issue can be solved, in some frames, the smoke is also counted as a part of the car. However, the number of affected frames are small (around 10 frames), making it barely visible by human eyes on a video. But still, it is a critical mistake, the further adjustment is needed.

As shown in Fig. 7, the smoke ratio of 1.4 solves the vanishing car issue perfectly. However, it still counts smoke as a car object in 1 to 2 frames out of a total of 140 frames (the video is about 5 seconds, so 1 to 2 frames are about  $\sim 0.1$  seconds). Also, in some frames, the front tire of car is recognized as a building object, but the rest of the car still segments properly. Since all the mistakes are barely visible by human eyes, it is our final choice for the whole project.

In summary, covering the car with a bigger area of smoke than the size of car is the best option to train the model.

## VI. DEHAZING USING MATLAB

We also use the “dehaze” function in Matlab to dehaze the smoke in the racing car video. As shown in Fig. 8. It improves the segment result significantly for the model trained with smoke ratio of 0.75. It does not work on model trained with smoke ratio of 1.4, and instead it makes performance worse.

The reason behind it remains unknown, but the experiment shows another option of using a combination of “dehazing + Unet + mobileone\_s4 model with smoke ratio of 0.75”. It could be useful in the further study and experiments.



Fig. 8: Dehazed frame with car fully covered by smoke (left) and the frame after dehazing and segmentation with model trained by smoke ratio 0.75 (right). The result has been significantly improved.

## VII. FUTURE WORK

### A. Car object identification and tire

It is worth noting that not all the pictures with smoke could disrupted the model. We are wondering what exactly caused the model failed to segment the car object covered by the smoke. After seeing  $\sim 100$  pictures with their smoke version and corresponding segmentation results, it is observed that the improper segmentation seems to be happening mostly on the tire, and then extend into the rest of car, especially when the image shows only the right/left side or rear of the car. It is also observed that a pile of tire is recognized as car object for most of the model combinations, stating that the tire could be an important feature for identifying a car object. Note that in our best model (Unet + mobileone with 1.4 smoke ratio), it sometimes identifies the tire as a building (which is not car) object. The reason why it has this issue but as well becomes the best overall performance still needs to be researched. Therefore, A research on how the presence of tire affect the segmentation of car object is recommended for further exploration.

### B. Non-smoke car vanishing issue

The “non-smoke car vanishing issue” is risen when the model fails to segment the car properly during the scenario where there is no smoke surrounding around the car. It seems to be happening only for the small dataset CamVid, and solved using a bigger dataset. The questions are:

- Whether we should consider the output that contains similar issue as a valid output.
- Whether we should continue on using this dataset even if we know that it is causing this issue.

A research on whether this issue would affect its validity of performance is needed to solve the above questions.

### C. Smoke and depth-wise separable convolution

The traditional convolution uses stride, which mixes up a few pixels together and obtain a gathered feature. However, such method may not be very useful on the smoke covered scenario. In my hypothesis, since a smoke covered area can have a very similar RGB values between adjacent pixels (even if these pixels belong to different objects). The use of stride would mix up these adjacent pixels’ RGB values, makes it



even harder to find the small RGB changes on adjacent pixels. But for depth-wise separable convolution that adapts the point-wise convolution. It will look through each pixel's R and G and B value separately, and then record it as a feature. By doing so, the difference between each pixel can be recorded within the feature and therefore easier to be distinguished later during the object segmentation. This could be a possible reason why deeper layer model like ResNet doesn't perform better than the shallower layer model like mobileone. A further research on comparing traditional convolution and depth-wise separable convolution on the smoke-covered objects can be conducted to test this hypothesis.

#### D. Heavy smoke

If the smoke tends to be really heavy, the R/G/B value of pixels will be greatly impacted. In that case, we can no longer consider the smoke as noise or a transparent object, but instead, as a non-transparent object. Which would certainly make the model unable to segment an object as it is completely blocked by the smoke. Such scenario is beyond the scope of object segmentation, and therefore not going to be discussed in this paper.

### VIII. CONCLUSION

The vanishing car issue can be solved by retraining a more suitable model with a more targeted (toward required scenario) dataset. In our attempt, we switch the original model with another model called "mobileone\_s4" that is more suitable and useful to solve the issue. Later on, we attempted to create a new dataset from the original dataset by adding smoke into the original dataset pictures. It is also found that expanding the original dataset by pictures with smoke is better than (on performance) replacing the dataset. We then tested it on a bigger dataset called cityscape with 2 different smoke ratio of 0.75 and 1.4. We found out that using the smoke ratio of 1.4 is the best option to solve the vanishing car issue while keeping the overall performance well. The dehazing function from Matlab is also tested in our experiment. It works well on the model trained by the smoke ratio of 0.75 and significantly improves the performance. While it does not improve our best model which is trained by the smoke ratio of 1.4, but instead makes its performance worse.

### REFERENCES

- [1] Howard, A. G., Zhu, M., Chen, B., Kalenichenko, D., Wang, W., Weyand, T., ... & Adam, H. (2017). Mobilenets: Efficient convolutional neural networks for mobile vision applications. *arXiv preprint arXiv:1704.04861*.
- [2] Chen, L. C., Papandreou, G., Schroff, F., & Adam, H. (2017). Rethinking atrous convolution for semantic image segmentation. *arXiv preprint arXiv:1706.05587*.
- [3] Everingham, M., Van Gool, L., Williams, C. K., Winn, J., & Zisserman, A. (2010). The pascal visual object classes (voc) challenge. *International journal of computer vision*, 88, 303-338.
- [4] M. Cordts, M. Omran, S. Ramos, T. Rehfeld, M. Enzweiler, R. Benenson, U. Franke, S. Roth, and B. Schiele, "The Cityscapes Dataset for Semantic Urban Scene Understanding," in *Proc. of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.
- [5] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).
- [6] Ronneberger, O., Fischer, P., & Brox, T. (2015). U-net: Convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention—MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18* (pp. 234-241). Springer International Publishing.
- [7] Radosavovic, I., Kosaraju, R. P., Girshick, R., He, K., & Dollár, P. (2020). Designing network design spaces. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 10428-10436).
- [8] Tan, M., & Le, Q. (2019, May). Efficientnet: Rethinking model scaling for convolutional neural networks. In *International conference on machine learning* (pp. 6105-6114). PMLR.
- [9] Vasu, P. K. A., Gabriel, J., Zhu, J., Tuzel, O., & Ranjan, A. (2023). Mobileone: An improved one millisecond mobile backbone. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 7907-7917).
- [10] Chaurasia, A., & Culurciello, E. (2017, December). Linknet: Exploiting encoder representations for efficient semantic segmentation. In *2017 IEEE visual communications and image processing (VCIP)* (pp. 1-4). IEEE.
- [11] Lin, T. Y., Dollár, P., Girshick, R., He, K., Hariharan, B., & Belongie, S. (2017). Feature pyramid networks for object detection. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 2117-2125).
- [12] Li, R., Zheng, S., Zhang, C., Duan, C., Su, J., Wang, L., & Atkinson, P. M. (2021). Multiattention network for semantic segmentation of fine-resolution remote sensing images. *IEEE Transactions on Geoscience and Remote Sensing*, 60, 1-13.